gravitational pull is embodied in the formula $g = Gm/r^2$, where r is distance from that bit of matter, m is the mass of that bit of matter, and G is Newton's gravitational constant. This G controls the overall strength of the gravitational pull.

In Einstein's more accurate, relativistic version of the gravitational laws, the strength of gravity, and the strength of all the warping of space and time produced by matter, are also proportional to this G.

If there is no bulk—if the only thing that exists is our four-dimensional universe—then Einstein's relativistic laws say that G is absolutely constant. The same everywhere in space. Never changing in time.

But if the bulk *does* exist, then the relativistic laws allow this G to change. It *might*, the Professor speculates, be controlled by bulk fields. It *probably is* controlled by bulk fields, he thinks. That's the best explanation for one of the observed anomalies (Figure 25.4) in my extrapolation of the movie's story.

The strength of the Earth's gravitational pull varies slightly from place to place due to the varying density of the rocks, oil, oceans, and atmosphere. Earth-orbiting satellites have mapped this varying strength. As of 2014 the most accurate map is from the European Space Agency's satellite GOCE[41] (top half of Figure 25.4). In 2014, the Earth's gravity is weakest in southern India (blue spot) and strongest in Iceland and Indonesia (red spots).